ICSE Computer Applications (Class 10) - Complete Chapter-wise MCQ Bank

Based on Previous Years' Papers (2008-2025)

CHAPTER 1: INTRODUCTION TO OBJECT ORIENTED PROGRAMMING CONCEPTS

Q1. A mechanism where one class acquires the properties of another class is called: [2023, 1 mark]

(a) Polymorphism
(b) Inheritance
(c) Encapsulation
(d) Abstraction

Answer: (b) Inheritance

Q2. Which of the following is not an OOP principle? [2015, 1 mark]

(a) Encapsulation
(b) Inheritance
(c) Polymorphism
(d) Compilation

Answer: (d) Compilation

Q3. Name any two basic principles of Object-oriented Programming: [2019, 2 marks]

(a) Abstraction and Inheritance
(b) Compilation and Execution

(c) Looping and Branching
(d) Input and Output

Answer: (a) Abstraction and Inheritance

Q4. "Java compiled code (byte code) can run on all operating systems" - Name this feature: [Specimen 2025, 1 mark]

(a) Robust and Secure
(b) Object Oriented
(c) Platform Independent
(d) Multithreaded

Answer: (c) Platform Independent

Q5. The feature of Java that allows the same method to have different implementations is called: [2014, 1 mark]

(a) Inheritance
(b) Polymorphism
(c) Encapsulation
(d) Abstraction

Answer: (b) Polymorphism

Q6. The wrapping of data and methods that operate on that data into a single unit is called: [Specimen 2020, 1 mark]

(a) Abstraction
(b) Inheritance
(c) Encapsulation
(d) Polymorphism

Answer: (c) Encapsulation

Q7. What is inheritance and how is it useful in Java? [2008, 2 marks]

(a) Process of hiding data; useful for security
(b) Process of acquiring properties of another class; useful for reusability
(c) Process of overloading methods; useful for flexibility
(d) Process of creating objects; useful for memory management

Answer: (b) Process of acquiring properties of another class; useful for reusability

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CHAPTER 2: ELEMENTARY CONCEPT OF OBJECTS AND CLASSES

Q1. What is an instance of a class called? [2023, 2 marks]

(a) Method
(b) Object
(c) Variable
(d) Constructor

Answer: (b) Object

Q2. Why is a class called a factory of objects? [2009, 2 marks]

(a) Because it produces objects
(b) Because it destroys objects
(c) Because it stores objects
(d) Because it compiles objects

Answer: (a) Because it produces objects

Q3. Why is class known as composite data type? [2009, 2 marks]

(a) It can store multiple primitive data types
(b) It requires a reference to use its attributes

(c) It is built into Java
(d) It cannot be modified

Answer: (b) It requires a reference to use its attributes

Q4. Employee staff = new Employee(); What does this statement mean? [2008, 2 marks]

(a) staff is a class, Employee is an object
(b) staff is an object of class Employee
(c) Employee is an object of class staff
(d) Both are classes

Answer: (b) staff is an object of class Employee

Q5. What is the use of the keyword 'new'? [Specimen 2020, 1 mark]

(a) To declare a variable
(b) To instantiate an object by allocating memory
(c) To define a class
(d) To import a package

Answer: (b) To instantiate an object by allocating memory

CHAPTER 3: VALUES AND DATA TYPES

Q1. Which of the following is a primitive data type? [2011, 1 mark]

(a) int
(b) String
(c) Array
(d) Class

Answer: (a) int

Q2. The size of 'n' (short data type) is: [Specimen 2025, 1 mark]

(a) 2 bytes
(b) 4 bytes
(c) 8 bytes
(d) 16 bytes

Answer: (a) 2 bytes

Q3. The default initial value of a boolean variable is: [2009, 2 marks]

(a) true
(b) false
(c) 0
(d) null

Answer: (b) false

Q4. The default value of a boolean variable is: [Specimen 2025, 1 mark]

(a) False
(b) 0
(c) false
(d) True

Answer: (c) false

Q5. Identify and name the following tokens: public, 'a', ==, { } [2019, 2 marks]

(a) Keyword, Literal, Operator, Separator
(b) Identifier, Character, Relational, Brace
(c) Modifier, String, Comparison, Block
(d) Access specifier, Constant, Equality, Curly

Answer: (a) Keyword, Literal, Operator, Separator

Q6. What are literals? [Specimen 2020, 1 mark]

(a) Variables that can change
(b) Constant values that can be assigned to variables
(c) Reserved words in Java
(d) Methods of a class

Answer: (b) Constant values that can be assigned to variables

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Q7. The number of bits occupied by the value 'a' (char) is: [2023, 1 mark]

(a) 1 bit
(b) 2 bits
(c) 4 bits
(d) 16 bits

Answer: (d) 16 bits

Q8. State the difference between a boolean literal and character literal: [2009, 2 marks]

(a) Boolean literal is true/false, character literal is a single character in quotes
(b) Both are same
(c) Boolean literal is numeric, character literal is alphabetic
(d) Boolean literal uses double quotes, character uses single

Answer: (a) Boolean literal is true/false, character literal is a single character in quotes

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CHAPTER 4: OPERATORS IN JAVA

Q1. Identify the type of operator &&: [2023, 1 mark]

(a) ternary
(b) unary

(c) logical
(d) relational

Answer: (c) logical

Q2. Identify the operator that gets the highest precedence while evaluating the expression: a + b % c * d - e [Specimen 2025, 1 mark]

(a) +
(b) %
(c) -
(d) *

Answer: (b) %

Q3. Which of the following is a valid java keyword? [Specimen 2025, 1 mark]

(a) If
(b) BOOLEAN
(c) static
(d) Switch

Answer: (c) static

Q4. Write a difference between unary and binary operator: [2019, 2 marks]

(a) Unary operates on one operand, binary on two operands
(b) Unary operates on two operands, binary on one
(c) Both operate on one operand
(d) Both operate on two operands

Answer: (a) Unary operates on one operand, binary on two operands

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Q5. Differentiate between = and == operators: [Specimen 2020, 1 mark]

(a) = is assignment, == is equality check
(b) = is equality check, == is assignment
(c) Both are same
(d) = is for strings, == is for numbers

Answer: (a) = is assignment, == is equality check

Q6. The output of 42 / 6 % 2 is: [2025, 1 mark]

(a) 1
(b) 10
(c) 2
(d) 0

Answer: (a) 1

Explanation: 42 / 6 = 7, 7 % 2 = 1

Q7. Evaluate the expression: b *= a++ - ++b + ++a when a=2, b=3: [Specimen 2025, 2 marks]

(a) a=4, b=6
(b) a=3, b=5
(c) a=4, b=5
(d) a=3, b=6

Answer: (a) a=4, b=6

Q8. The statement used to increase the value of variable x by 2 is: [Specimen 2025, 1 mark]

(a) x += 2
(b) x = 2
(c) x = x + 2
(d) Both A and C

Answer: (d) Both A and C

CHAPTER 5: INPUT IN JAVA

Q1. The Scanner class method used to accept words with space is: [2023, 1 mark]

(a) next()
(b) nextLine()
(c) nextWord()
(d) nextString()

Answer: (b) nextLine()

Q2. Which package contains the Scanner class? [2011, 1 mark]

(a) java.io
(b) java.util
(c) java.lang
(d) java.awt

Answer: (b) java.util

Q3. The method to convert a string to a primitive integer data type is: [2009, 2 marks]

(a) Integer.parseInt()
(b) Integer.toString()
(c) String.toInt()
(d) Integer.valueOf()

Answer: (a) Integer.parseInt()

Q4. Which of the following converts "25" to 25.0? [2025, 1 mark]

(a) Double.parseDouble("25")
(b) Double.parse("25")

(c) Double.parseDouble(25)
(d) Double.parseDouble("25")

Answer: (a) Double.parseDouble("25")

Q5. Write a Java statement to input a character from the user using the keyboard: [2008, 2 marks]

(a) char ch = sc.nextChar();
(b) char ch = sc.nextLine().charAt(0);
(c) char ch = sc.readChar();
(d) char ch = sc.getChar();

Answer: (b) char ch = sc.nextLine().charAt(0);

Q6. Missing a semicolon in a statement is what type of error? [2023, 1 mark]

(a) Logical
(b) Syntax
(c) Runtime
(d) No error

Answer: (b) Syntax

Q7. Differentiate between next() and nextLine() methods: [2017, 2 marks]

(a) next() reads a word, nextLine() reads a line
(b) next() reads a line, nextLine() reads a word
(c) Both read the same
(d) next() reads numbers, nextLine() reads strings

Answer: (a) next() reads a word, nextLine() reads a line

CHAPTER 6: MATHEMATICAL LIBRARY METHODS

Q1. What value will Math.sqrt(Math.ceil(15.3)) return? [2023, 1 mark]

(a) 16.0
(b) 16
(c) 4.0
(d) 5.0

Answer: (c) 4.0

Explanation: Math.ceil(15.3) = 16.0, Math.sqrt(16.0) = 4.0

Q2. The output of Math.max(-7, Math.min(-5, -4)) is: [2025, 1 mark]

(a) -5
(b) -4
(c) -7
(d) error

Answer: (a) -5

Explanation: Math.min(-5,-4) = -5, Math.max(-7,-5) = -5

Q3. The output of Math.ceil(6.4) + Math.floor(-1-2) is: [Specimen 2025, 1 mark]

(a) 3.0
(b) 4
(c) 3
(d) 4.0

Answer: (d) 4.0

Explanation: Math.ceil(6.4)=7.0, Math.floor(-3)= -3.0, 7.0 + (-3.0) = 4.0

Q4. Write a Java expression for

x3+y

3

x

$+$

y

: [Specimen 2025, 2 marks]

(a) Math.cbrt(x) + Math.sqrt(y)
(b) Math.pow(x,1/3) + Math.pow(y,1/2)
(c) Math.cuberoot(x) + Math.square(y)
(d) Both A and B

Answer: (d) Both A and B

Q5. Write a Java expression for

(a+b)n3+b

3

$+b$

$(a+b)$

n

: [2009, 2 marks]

(a) Math.pow((a+b),n)/(Math.sqrt(3)+b)
(b) Math.pow(a+b,n)/(Math.sqrt(3)+b)
(c) (a+b)^n/(Math.sqrt(3)+b)
(d) Math.pow(a+b,n)/(Math.sqrt(3)+b)

Answer: (d) Math.pow(a+b,n)/(Math.sqrt(3)+b)

Q6. Write a Java expression for (a+b)^x: [2023, 2 marks]

(a) Math.pow(a+b, x)
(b) Math.pow(a,b) + x
(c) Math.power(a+b, x)
(d) (a+b)^x

Answer: (a) Math.pow(a+b, x)

Q7. Give the output of Math.pow(25,0.5) + Math.ceil(4.2): [Specimen 2020, 2 marks]

(a) 10.0
(b) 9.0
(c) 10
(d) 9

Answer: (a) 10.0

CHAPTER 7: CONDITIONAL CONSTRUCTS IN JAVA

Q1. The absence of which statement leads to fall through situation in switch case? [2023, 1 mark]

(a) continue
(b) break
(c) return
(d) System.exit(0)

Answer: (b) break

Q2. Which of the following is not true with regards to a switch statement? [Specimen 2025, 1 mark]

(a) checks for equality between input and case labels
(b) supports floating point constants
(c) break is used to exit from switch block
(d) case labels are unique

Answer: (b) supports floating point constants

Q3. Rewrite the following code using single if statement: if(code=='g') System.out.println("GREEN"); else if(code=='G') System.out.println("GREEN"); [Specimen 2025, 2 marks]

(a) if(code=='g' || code=='G') System.out.println("GREEN");
(b) if(code=='g' && code=='G') System.out.println("GREEN");
(c) if(code=='g') System.out.println("GREEN");
(d) if(code!='g' || code!='G') System.out.println("GREEN");

Answer: (a) if(code=='g' || code=='G') System.out.println("GREEN");

Q4. Differentiate between if else if and switch-case statements: [2019, 2 marks]

(a) if-else can test boolean expressions, switch tests equality
(b) if-else is faster, switch is slower
(c) if-else uses break, switch does not
(d) Both are same

Answer: (a) if-else can test boolean expressions, switch tests equality

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Q5. What is the output when n=10? switch(n) { case 10: System.out.print(n*2); case 4: System.out.print(n*4); break; default: System.out.print(n); } [2023, 1 mark]

(a) 20
(b) 10
(c) 2040
(d) 10

Answer: (c) 2040

CHAPTER 8: ITERATIVE CONSTRUCTS IN JAVA

Q1. State the type of loop in: for(int i=5; i=0; i-=2) System.out.println(i); [2023, 1 mark]

(a) finite
(b) infinite
(c) null
(d) fixed

Answer: (b) infinite

Q2. Select the infinite loop: [2025, 1 mark]

(a) for(int i=1; i<=10; i++)
(b) for(int i=2; i!=0; i-=3)
(c) for(int i=5; i<=5; i++)
(d) for(int i=1; i>=1; i--)

Answer: (b) for(int i=2; i!=0; i-=3)

Q3. To execute a loop 10 times, which of the following is correct? [Specimen 2025, 1 mark]

(a) for(int i=11; i<=30; i+=2)
(b) for(int i=11; i<=30; i+=3)
(c) for(int i=11; i<20; i++)
(d) for(int i=11; i<=21; i++)

Answer: (a) for(int i=11; i<=30; i+=2)

Q4. How many times will the following loop execute? int x=10; while(true) { System.out.println(x++ *2); if(x%3==0) break; } [Specimen 2025, 2 marks]

(a) 2 times
(b) 3 times
(c) 4 times
(d) Infinite

Answer: (a) 2 times

Q5. Convert the following do...while loop to for loop: int x=10; do { x--; System.out.print(x); } while(x>=1); [2023, 2 marks]

(a) for(int x=9; x>=1; x--) System.out.print(x);
(b) for(int x=10; x>=1; x--) System.out.print(x);
(c) for(int x=9; x>=0; x--) System.out.print(x);
(d) for(int x=10; x>=0; x--) System.out.print(x);

Answer: (a) for(int x=9; x>=1; x--) System.out.print(x);

Q6. Convert the given while loop into an equivalent do while loop: int x=10,c=20; while(c>=10) { x++; c=c-2; } [2009, 3 marks]

(a) do { x++; c=c-2; } while(c>=10);
(b) do { x++; c=c-2; } while(c>10);
(c) do { x++; c=c-2; } while(c<=10);
(d) do { x++; c=c-2; } while(c==10);

Answer: (a) do { x++; c=c-2; } while(c>=10);

CHAPTER 9: NESTED FOR LOOPS

Q1. A physical education teacher asks the students to do the side stretch 10 times. Which programming construct does the teacher use? [2025, 1 mark]

(a) if
(b) switch

(c) for
(d) if else if

Answer: (c) for

Q2. A clock is a real-life example of nested loops because: [2025, 1 mark]

(a) hour hand moves through 12 positions, minute hand through 60 positions within each hour
(b) both hands move simultaneously
(c) only hour hand moves
(d) only minute hand moves

Answer: (a) hour hand moves through 12 positions, minute hand through 60 positions within each hour

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Q3. Write a program to display the following pattern using nested loops: 1 22 333 4444 55555. The correct nested loop structure is: [2009, 15 marks]

(a) for(i=1; i<=5; i++) { for(j=1; j<=i; j++) System.out.print(i); }
(b) for(i=1; i<=5; i++) { for(j=1; j<=5; j++) System.out.print(i); }
(c) for(i=1; i<=5; i++) { for(j=1; j<=i; j++) System.out.print(j); }
(d) for(i=1; i<=5; i++) { for(j=1; j<=5; j++) System.out.print(j); }

Answer: (a) for(i=1; i<=5; i++) { for(j=1; j<=i; j++) System.out.print(i); }

CHAPTER 10: CLASS AS THE BASIS OF ALL COMPUTATION

Q1. An instance of a class is called: [2023, 2 marks]

(a) Object
(b) Method
(c) Variable
(d) Constructor

Answer: (a) Object

Q2. The method which has the same name as that of the class name is called: [2023, 2 marks]

(a) Constructor
(b) Method
(c) Function
(d) Procedure

Answer: (a) Constructor

Q3. What is the use of the keyword 'this'? [2009, 2 marks]

(a) To refer to the current object
(b) To create a new object
(c) To destroy an object
(d) To import a class

Answer: (a) To refer to the current object

Q4. Differentiate between instance variable and class variable: [2011, 2 marks]

(a) Instance variable belongs to object, class variable belongs to class
(b) Both belong to object
(c) Both belong to class
(d) Instance variable is static, class variable is non-static

Answer: (a) Instance variable belongs to object, class variable belongs to class

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CHAPTER 11: USER-DEFINED METHODS

Q1. Write the prototype of a function search which takes two arguments a string and a character and returns an integer value: [Specimen 2020, 1 mark]

(a) int search(String str, char ch)
(b) void search(String str, char ch)
(c) search(String str, char ch)
(d) int search(String, char)

Answer: (a) int search(String str, char ch)

Q2. Write a method prototype named check() which takes an integer argument and returns a char: [2023, 1 mark]

(a) char check()
(b) void check(int x)
(c) check(int x)
(d) char check(int x)

Answer: (d) char check(int x)

Q3. The number of values that a method can return is: [2023, 1 mark]

(a) 1
(b) 2
(c) 3
(d) 4

Answer: (a) 1

Q4. The statement that brings the control back to the calling method is: [Specimen 2025, 1 mark]

(a) break
(b) System.exit(0)
(c) continue
(d) return

Answer: (d) return

Q5. Difference between pure and impure functions: [2009, 2 marks]

(a) Pure returns value, impure does not return value
(b) Pure modifies state, impure does not
(c) Both return value
(d) Both do not return value

Answer: (a) Pure returns value, impure does not return value

Q6. Differentiate between call by value and call by reference: [2019, 2 marks]

(a) Call by value copies value, call by reference passes reference
(b) Both copy value
(c) Both pass reference
(d) Call by value is faster

Answer: (a) Call by value copies value, call by reference passes reference

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CHAPTER 12: CONSTRUCTORS

Q1. The method which has the same name as that of the class name is called: [2023, 2 marks]

(a) Constructor
(b) Destructor
(c) Method
(d) Function

Answer: (a) Constructor

Q2. Name the two types of constructors: [2015, 1 mark]

(a) Parameterized and Non-Parameterized
(b) Default and Copy
(c) Static and Non-static
(d) Public and Private

Answer: (a) Parameterized and Non-Parameterized

Q3. Write two characteristics of a constructor: [2018, 2 marks]

(a) Same name as class, no return type
(b) Different name from class, has return type
(c) Same name as class, has return type
(d) Different name from class, no return type

Answer: (a) Same name as class, no return type

Q4. What is constructor overloading? [Specimen 2020, 1 mark]

(a) Having multiple constructors with different parameters
(b) Having one constructor with multiple parameters
(c) Having multiple classes with same constructor
(d) Having constructor with return type

Answer: (a) Having multiple constructors with different parameters

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Q5. Name the type of constructors used in the program: calculate() { a=0; b=0.0; } calculate(int x, double y) { a=x; b=y; } [Specimen 2025, 2 marks]

(a) Default and Parameterized constructor
(b) Copy and Parameterized constructor
(c) Static and Non-static constructor
(d) Private and Public constructor

Answer: (a) Default and Parameterized constructor

CHAPTER 13: LIBRARY CLASSES (WRAPPER CLASSES)

Q1. Character class methods are found in the package called: [2025, 1 mark]

(a) java.util
(b) java.lang
(c) java.awt
(d) java.io

Answer: (b) java.lang

Q2. The method that converts a character to uppercase is: [2022 Sem 2, 1 mark]

(a) toUpper()
(b) ToUpperCase()
(c) toUppercase()
(d) toUpperCase(char)

Answer: (d) toUpperCase(char)

Q3. The method to convert a lowercase character to uppercase is: [Specimen 2025, 1 mark]

(a) String.toUpperCase()
(b) Character.isUpperCase(char)
(c) Character.toUpperCase(char)
(d) toUpperCase()

Answer: (c) Character.toUpperCase(char)

Q4. Boxing is the conversion of: [Specimen 2020, 1 mark]

(a) Primitive to Wrapper object
(b) Wrapper object to Primitive

(c) String to Integer
(d) Integer to String

Answer: (a) Primitive to Wrapper object

Q5. The corresponding wrapper class of float data type is: [2022 Sem 2, 1 mark]

(a) FLOAT
(b) float
(c) Float
(d) Floating

Answer: (c) Float

Q6. Class used to convert a primitive data type to its corresponding object is: [2022 Sem 2, 1 mark]

(a) String
(b) Wrapper
(c) System
(d) Math

Answer: (b) Wrapper

Q7. What is unboxing? [Specimen 2020, 1 mark]

(a) Converting wrapper object to primitive
(b) Converting primitive to wrapper object
(c) Converting string to integer
(d) Converting integer to string

Answer: (a) Converting wrapper object to primitive

CHAPTER 14: ENCAPSULATION (ACCESS SPECIFIERS)

Q1. Mention the types of access specifiers: [Specimen 2020, 1 mark]

(a) public, private, protected, default
(b) public, private, static, final
(c) public, private, abstract, final
(d) public, private, static, default

Answer: (a) public, private, protected, default

Q2. The access modifier that gives least accessibility is: [2022 Sem 2, 1 mark]

(a) private
(b) public
(c) protected
(d) package

Answer: (a) private

Q3. The access specifier that gives the most accessibility is: [2015, 1 mark]

(a) public
(b) private
(c) protected
(d) default

Answer: (a) public

Q4. Post office is an example of which access specifier? [2025, 1 mark]

(a) public
(b) local
(c) protected
(d) private

Answer: (a) public

Q5. How are private members of a class different from public members? [2018, 2 marks]

(a) Private accessible only within class, public accessible everywhere
(b) Private accessible everywhere, public only within class
(c) Both accessible everywhere
(d) Both accessible only within class

Answer: (a) Private accessible only within class, public accessible everywhere

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CHAPTER 15: ARRAYS

Q1. Consider the two-dimensional array P[2][3] of peripherals. State the index of the device Barcode Scanner: [2025, 1 mark]

(a) P[1][1]
(b) P[0][1]
(c) P[1][2]
(d) P[0][0]

Answer: (a) P[1][1]

Q2. The index (subscript) of the last element of an array 'ar' is: [2025, 1 mark]

(a) ar.length()
(b) ar[]length
(c) ar.length() - 1
(d) ar.length - 1

Answer: (d) ar.length - 1

Q3. A single dimensional array has 50 elements. The correct statement to initialize the last element to 100 is: [Specimen 2025, 1 mark]

(a) x[51] = 100
(b) x[48] = 100
(c) x[49] = 100
(d) x[50] = 100

Answer: (c) x[49] = 100

Q4. The number of values that a method can return is: [2023, 1 mark]

(a) 1
(b) 2
(c) 3
(d) 4

Answer: (a) 1

Q5. The statement used to find the total number of Strings present in the string array String s[] is: [Specimen 2025, 1 mark]

(a) s.length
(b) s.length()
(c) length(s)
(d) len(s)

Answer: (a) s.length

Q6. What is the output of System.out.print(a[0] + a[5]); for int a[] = {12,10,8,4,6,2,3,5,7}: [2025, 2 marks]

(a) 14
(b) 12
(c) 10
(d) 8

Answer: (a) 14

Q7. For the same array, the index of the largest element is: [2025, 2 marks]

(a) 0
(b) 1
(c) 2
(d) 3

Answer: (a) 0

Q8. The position of 34 in int x[][] = {{2,4,5,6},{5,7,8,1},{34,1,10,9}} is: [Specimen 2025, 2 marks]

(a) x[2][0]
(b) x[0][2]
(c) x[1][0]
(d) x[2][1]

Answer: (a) x[2][0]

Q9. The result of x[2][3] + x[1][2] for the same array is: [Specimen 2025, 2 marks]

(a) 17
(b) 15
(c) 18
(d) 16

Answer: (a) 17

CHAPTER 16: STRING HANDLING

Q1. The String class method to join two strings is: [2023, 1 mark]

(a) concat(String)
(b) <string>.join(String)
(c) concat(char)
(d) Concat()

Answer: (a) concat(String)

Q2. The output of "COMPOSITION".substring(3,6) is: [2023, 1 mark]

(a) POSI
(b) POS
(c) MPO
(d) MPOS

Answer: (b) POS

Q3. Predict the output: String P="20", Q="22"; int a = Integer.parseInt(P); int b = Integer.valueOf(Q); System.out.println(a+" "+b); [2023, 1 mark]

(a) 20
(b) 20 22
(c) 2220
(d) 22

Answer: (b) 20 22

Q4. The output of "SUCCESS".indexOf('S') + "SUCCESS".lastIndexOf('S') is: [2022 Sem 2, 1 mark]

(a) 0
(b) 5
(c) 6
(d) -5

Answer: (c) 6

Q5. The output of "Good".concat("Day") is: [2022 Sem 2, 1 mark]

(a) GoodDay
(b) Good Day
(c) Goodday
(d) goodDay

Answer: (a) GoodDay

Q6. The output of "Lucknow".substring(0,4) is: [2022 Sem 2, 1 mark]

(a) Lucknow
(b) Luckn
(c) Luck
(d) luck

Answer: (c) Luck

Q7. The output of System.out.println("I said, "It's wise to obey elders.""); is: [2025, 1 mark]

(a) I said, 'It is wise to obey elders.'
(b) I said, "It's wise to obey elders."
(c) I said, 'It's wise to elders.'
(d) "It's wise to obey elders."

Answer: (b) I said, "It's wise to obey elders."

Q8. The output of "ANGER".compareTo("ANGEL") is: [2025, 1 mark]

(a) 3
(b) -6
(c) 6
(d) 0

Answer: (c) 6

Q9. The output of System.out.println(x.substring(3).concat(y.substring(3))) for x="talent", y="matrix" is: [2025, 2 marks]

(a) entrix
(b) entritx
(c) talrix
(d) talentrix

Answer: (a) entrix

Q10. Write a Java statement to extract the second character of the string str: [Specimen 2020, 2 marks]

(a) str.charAt(1)
(b) str.charAt(2)
(c) str.charAt(0)
(d) str.charAt(str.length()-1)

Answer: (a) str.charAt(1)

Q11. The method that returns a String is: [Specimen 2025, 1 mark]

(a) length()
(b) charAt(int)
(c) replace(char, char)
(d) indexOf(String)

Answer: (c) replace(char, char)

CHAPTER 17: SORTING AND SEARCHING

Q1. What is the difference between linear search and binary search? [2019, 2 marks]

(a) Linear works on sorted/unsorted, binary works on sorted only
(b) Linear works on sorted only, binary works on unsorted only
(c) Both work on sorted only
(d) Both work on unsorted only

Answer: (a) Linear works on sorted/unsorted, binary works on sorted only

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Q2. The search or sort algorithm that makes several passes through the array, selecting the next smallest item each time is: [2012, 2 marks]

(a) Selection Sort
(b) Binary Search
(c) Bubble Sort
(d) Linear Search

Answer: (a) Selection Sort

Q3. The search or sort algorithm that at each stage compares the sought key value with the middle element is: [2012, 2 marks]

(a) Binary Search
(b) Linear Search
(c) Bubble Sort
(d) Selection Sort

Answer: (a) Binary Search

CHAPTER 18: METHOD OVERLOADING

Q1. When there are multiple definitions with the same function name, what makes them different? [2009, 2 marks]

(a) Parameter list
(b) Return type

(c) Access specifier
(d) Method body

Answer: (a) Parameter list

Q2. Which pair of methods will cause a compile-time error due to incorrect method overloading? [2025, 1 mark]

(a) void test(int a, int b) and void test(double a, double b)
(b) void test(int a, double b) and void test(double a, int b)
(c) void test(int a, double b) and void test(int a)
(d) void test(int a) and int test(int a)

Answer: (d) void test(int a) and int test(int a)

Q3. What is function overloading? [2012, 2 marks]

(a) Multiple functions with same name but different parameters
(b) Multiple functions with different names
(c) Single function with multiple parameters
(d) Function calling another function

Answer: (a) Multiple functions with same name but different parameters

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CHAPTER 19: EXCEPTION HANDLING

Q1. What is an exception? [2010, 1 mark]

(a) An abnormal condition that occurs during program execution
(b) A syntax error
(c) A logical error
(d) A compilation error

Answer: (a) An abnormal condition that occurs during program execution

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Q2. Name two exception handling blocks: [2010, 1 mark]

(a) try and catch
(b) if and else
(c) for and while
(d) class and object

Answer: (a) try and catch

Q3. State the type of errors in the following: (i) 100/0 (ii) Missing semicolon: [2016, 2 marks]

(a) Runtime error, Syntax error
(b) Syntax error, Runtime error
(c) Logical error, Syntax error
(d) Runtime error, Logical error

Answer: (a) Runtime error, Syntax error